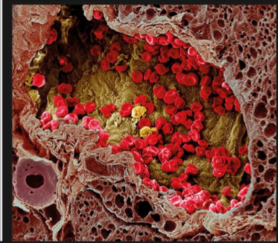
IMAGING THE BODY

IMAGING IS A VITAL PART OF DIAGNOSING ILLNESS, UNDERSTANDING DISEASE, AND EVALUATING TREATMENTS. MODERN TECHNIQUES PROVIDING HIGHLY DETAILED INFORMATION HAVE LARGELY REPLACED SURGERY AS A METHOD OF INVESTIGATION.

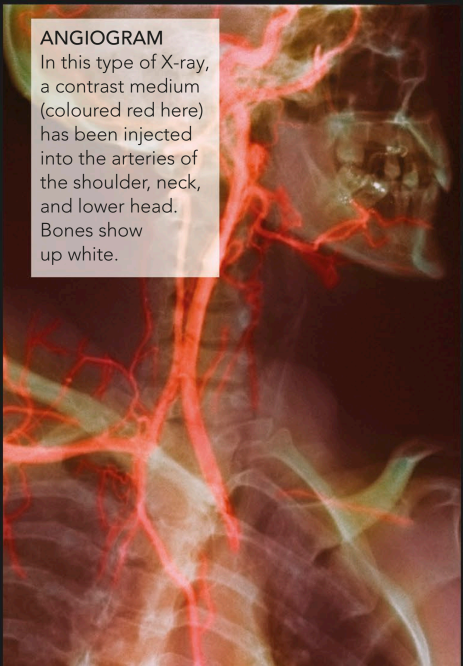
The invention of the X-ray made the development of non-invasive medicine possible. Without the ability to see inside the body, many disorders could be found only after major surgery. Computerized imaging now helps doctors make early diagnoses, often greatly increasing the chances of recovery. Computers process and enhance raw data, for example re-interpreting shades of grey from an X-ray or scan into colours. However, sometimes direct observation is essential. Viewing techniques have also become less invasive with the development of instruments such as the endoscope (see opposite). This book makes extensive use of internal images from real bodies.

MICROSCOPY

In light microscopy (LM), light is passed through a section of material and lenses magnify the view up to 2,000 times. Even higher magnifications are possible with scanning electron microscopy (SEM), in which light runs across a specimen coated with gold film. Electrons bounce off the surface, creating a three-dimensional image.



SEM OF TUMOUR BLOOD SUPPLY
This image, in which the specimen has been frozen and split open, shows a blood vessel with blood cells growing into a melanoma (skin tumour).



ANGIOGRAM
In this type of X-ray, a contrast medium (coloured red here) has been injected into the arteries of the shoulder, neck, and lower head. Bones show up white.

X-RAY

X-rays are similar to light waves, but of very short wavelength. When passed through the body they create shadow images on photographic film. Dense structures such as bone show up white; soft tissues appear as shades of grey. To show up hollow or fluid-filled structures, these are filled with a substance that absorbs X-rays (a contrast medium). Fluoroscopy uses X-rays to gain real-time moving images of body parts, for instance to investigate swallowing.

X-RAY OF THE BREAST
A plain X-ray of the female breast (mammogram) is used as a routine screening test for breast cancer, which may show up as an unusually white area. This mammogram shows a healthy breast.